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An application of an improved FCOS algorithm in detection and recognition of industrial instruments

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Abstract

In order to transform the digital workshop, this paper adopts an improved FCOS algorithm to detect the digital display area of industrial instruments and recognize the numbers in the workshop. Specifically, based on the FCOS basic network model, the Balanced Feature Pyramid is combined with FPN to further integrate the feature information and enhance the original feature. The Softer-NMS algorithm instead of the traditional non maximum suppression algorithm, improves the detection accuracy. The experiment shows that the trained network model can be applied to the detection and recognition of industrial instruments, and it has certain validity and accuracy.

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Keywords: *Digital workshop; Industrial instrument detection and identification; FCOS; FPN; Softer - NMS;*

1. The introduction

In recent years, Industry 4.0 in Germany, industrial Internet in the United States and intelligent manufacturing 2025 in China have been successively proposed, making the manufacturing industry gradually networked and intelligent. As the core unit of intelligent manufacturing, digital workshop has started the development road of digital transformation and digital innovation. The digital transformation of the traditional workshop has become a hot issue nowadays. There are many industrial instruments in the digital workshop. Due to the limitation of working conditions and the requirement of economy and convenience, some old industrial instruments have no peripheral

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data communication interface. How to collect the data of these instruments has become a big problem. At the same time, the problem can be extended to the digital detection and identification of scenes including industrial instruments, which has great research value and practical demand. Traditional method has many disadvantages such as high labor costs, low efficiency and high error faults. In some industrial occasions, such as high temperature and high pressure, low temperature and humidity, there is a big risk and feasibility. But machine vision can ignore these unfavorable factors. It is applied to the poor working environment during a long period of time. It can not only provide convenience for industrial instruments' number to read, but also can improve the work efficiency.

The detection and recognition of industrial instruments can also be said to be automatic meter reading technology. This technology began to rise in the 1980s and began to develop rapidly after the 1990s. Digital recognition has always been a major direction in the field of text recognition. At present, many scientific and technical personnel at home and abroad have done a lot of in-depth research on digital recognition based on machine vision. The mainstream methods include traditional template matching methods, machine learning methods, and deep learning methods. Traditional recognition methods are algorithms based on image preprocessing and character segmentation. The main steps include: image acquisition, image preprocessing, character segmentation, and character recognition. The character segmentation adopts MSER (Maximally Stable Extremal Regions) segmentation method, and the recognition method adopts threading method or template matching method. The automatic meter reading recognition algorithm based on the image recognition algorithm uses the characteristics of the digits for recognition. Because the characteristics of the digits are very susceptible to environmental factors, they are not uniform in large changes in illumination, large background noise, and instrument tilt, picture resolution, poor quality and other extreme conditions, resulting in low recognition accuracy. There are problems with cumbersome image preprocessing, poor generalization ability, and can only be used in a small range and fixed scenes. For example, a camera is installed at a fixed position to shoot a piece of instruments, ensuring that the camera's shooting plane is parallel to the plane of the instruments' digital display area, but this will increase the installation cost of the camera and reduce the utilization rate of the camera. At the same time, due to the harsh environment in the factory, the service life of the camera is greatly reduced. Another mainstream recognition method is to use additional feature extraction methods and then use machine learning methods for training and learning to obtain results. Since machine learning methods rely on additional feature extraction work, the features extracted by different methods are not the same, which will have a certain impact on the recognition results and have poor adaptability. Therefore, deep learning methods have begun to be widely used to solve such problems.

Compared with machine learning methods that rely on additional extraction of features, deep learning methods can use multi-layer neural networks to process images, and then extract a variety of complex features to obtain more accurate detection and recognition results. In recent years, deep learning methods have been widely used in machine vision, speech recognition, text detection and other fields. For digital text detection and recognition problems in a non-single fixed environment, deep learning is significantly better than traditional methods and machine learning method. Therefore, in order to solve this problem, with the help of deep learning target detection algorithm, both the detection task and the recognition task can be realized. Common target detection algorithms are divided into two-stage target detection algorithms and single-stage target detection algorithms^{1,2,3}, and the FCOS⁴ algorithm based on anchor free. The FCOS algorithm is compared with previous targets Detection algorithms, such as Fast R-CNN⁵, Faster R-CNN⁶, YOLO v3⁷, Mask RCNN⁸etc., have obvious advantages. They do not need to rely on predefined anchor boxes, avoid complex calculations related to anchor boxes (such as IOU calculations), reduce computing memory, and greatly improve the detection efficiency. The operation efficiency far exceeds YOLO⁴.

Therefore, this paper solves the problem of detection and recognition of industrial instrument readings in the workshop, that is, first uses the FCOS⁴algorithm to detect the digital display area of the industrial instruments, and then recognizes the readings to improve the recognition efficiency and accuracy, and effectively reduce the false detection rate and leakage.

2. The process of digital detection and recognition of industrial instruments

During the process of the production, the angles and backgrounds of the industrial instruments may vary. Some samples captured are shown in Figure 1. The digital detection and recognition algorithm based on improved FCOS⁴

in this paper has certain generalization ability and can adapt to various problems caused by extreme environmental conditions.



Fig. 1. Industrial instrument diagram taken in practice.

The digital detection and recognition process of industrial instruments in this paper is shown in Figure 2. The image preprocessing of the industrial instrumentation data set in the experiment is carried out. Because the target detection area is small, it is susceptible to interference from light and other factors. Technologies such as median filtering, image binarization, image enhancement, etc., remove irrelevant information in the image and reduce the interference of external factors on the detection and recognition results. These technologies effectively improve the accuracy of detection and recognition. The pre-processed images were divided into training set, test set and verification set according to the ratio of 8:1:1, and then input into the FCOS⁴ target detection network. The training set is used for the training of classification model, and the classification including industrial instruments' digital display region and 0-9 numbers is established. After the model training is completed, the model parameters are adjusted through the verification set. Finally, the test set is used to test the model, evaluate the final detection results of the model, and output the classification and location of corresponding types. It classifies all the digital region types into one category and names them Number in this paper. Thus, the complexity of the model is reduced and the accuracy of the model is improved. And then identify them.

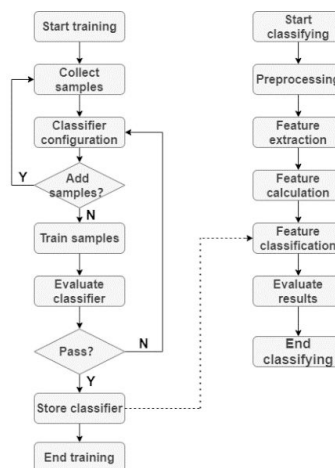


Fig. 2. The flow chart.

3. Detection and recognition of industrial instruments

3.1. FCOS (Fully Convolutional One-Stage Object Detection)

The algorithm in this paper adopts a single-stage target detection strategy. For the positioning and detection of the industrial instruments' reading region, the FCOS⁴ algorithm is used and improved on this basis, so that the algorithm has a good detection and positioning effect and high efficiency and accuracy for this detection task.

Tian Z et al. proposed a new target detection method FCOS¹ in 2019, and its network structure is shown in Figure 3. It is mainly divided into four steps. After inputting the image, the image is preprocessed first, and then the C3, C4, and C5 layers are extracted through the feature extraction backbone network (the out stride of three layers are 8, 16, 32 respectively). The feature extraction backbone network is The Resnet network. Then use the FPN structure to obtain five prediction layers: P3, P4, P5, P6, and P7. The P3, P4, and P5 layers are obtained through a 1×1 convolution of the C3, C4, and C5 layers, and the P6, P7 is obtained by down-sampling, and five feature maps with different scales are obtained, which are $1/128$, $1/64$, $1/32$, $1/16$, $1/8$ of the input image, and then make predictions on each prediction layer, that is to say, perform pixel-by-pixel regression on these 5 feature maps. If the prediction is only made in the last layer of the feature extraction network, there is very little position information. In order to avoid this problem, the FPN network is used.

Since there will be a lot of prediction results on the 5 feature maps in the end, this will increase the recall rate, but at the same time there will also be many prediction boxes with lower quality and far from the truth box. In order to reduce the occurrence of this problem, Adding the Center-ness layer is equivalent to adding a branch in the network prediction, parallel to the classification branch, to suppress those invalid prediction boxes. Many low-quality prediction frames can be directly filtered out through the scores predicted by the network, and then filtered by the NMS algorithm, a prediction frame that is very close to the truth frame can be obtained.

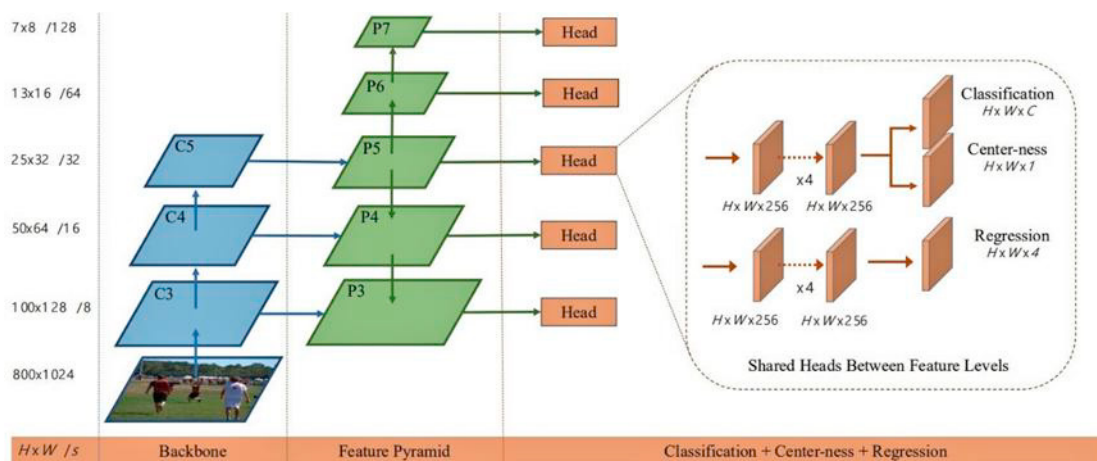


Fig. 3. Figure of FCOS network structure⁷.

3.2. FPN

Combining Balanced Feature Pyramid⁹ with FPN^{10,11} so that FPN^{10,11} features can be used more efficiently. This process is mainly to enhance the characteristics of different levels through the information obtained after the same depth of fusion and balance. It mainly includes four steps. Rescaling the multi-layer features of the P3 and P5 layers obtained by FPN to the size of P4, and then perform Integrate, calculate the weighted average value of the feature P. The calculation of the feature P is given by formula (1), and then refine the returns to P3 and P5. Through rescale, integrate and refine, strengthening can further integrate feature information, and finally add it to the original feature, so that the balanced feature can further enhance the original feature.

$$P = \frac{1}{L} \sum_{l=3}^{l=5} P_l \quad (1)$$

P6 and P7 are still obtained through up-sampling, and the fused new features are used in subsequent target detection, and the process is the same as FPN. The whole process is shown in Figure 4.

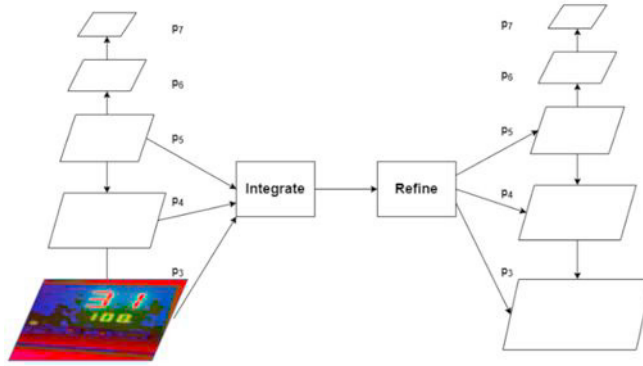


Fig. 4. The process of Balanced Feature Pyramid.

3.3. Softer-NMS

The original algorithm uses NMS¹² for the only post-processing, but if the intersection of the two candidate frame regions is relatively high, the original NMS will suppress the prediction frame with a lower confidence level, resulting in missed detection of a certain area, thereby the missing detection rate is greatly improved. To solve this problem, Soft-NMS¹³ attenuates the score of candidate frames whose IOU is greater than the threshold instead of directly deleting them. Softer-NMS takes the weighted average of all the coordinates of the detection frame whose intersection ratio of the detection frame with the highest classification confidence is greater than a certain threshold as the detection result. Therefore, this paper improves the original NMS to use Softer-NMS, and adopts a new minimized prediction box regression loss function (KL Loss). KL Loss is the Gaussian distribution of the prediction box and the KL dispersion of the Dirac distribution of the real box. The loss function of degree is calculated by formulas (2) – (4):

$$P_\theta(x) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x-x_e)^2}{2\sigma^2}} \quad (2)$$

$$P_D(x) = \delta(x - x_g) \quad (3)$$

$$\text{Loss}_{\text{reg}} = D_{\text{KL}}(P_\theta(x) \| P_D(x)) \quad (4)$$

Among them, $P_\theta(x)$ is the Gaussian distribution of the predicted frame; $P_D(x)$ ¹⁴ is the Dirac distribution of the real frame; x_e is the expected position of the predicted frame and x_g is the position of the real frame. Because $P_D(x)$ is the limit of a Gaussian distribution whose standard deviation is close to 0, so the standard deviation of $P_\theta(x)$ is close to 0, and Loss_{reg} will also be close to 0, representing the prediction box and the closer the real frame is, the standard deviation of the predicted frame can be used as the positional reliability to effectively locate the predicted frame.

The threshold is set to 0.5 in this paper. If $\text{IOU} > 0.5$, it is considered a correct detection, otherwise it is considered a wrong detection.

3.4. Industrial instrument digital recognition

We save the candidate area located in the digital display region of the instrument as a new data set. Hough correction is used to perform tilt correction operation on the picture in order to obtain a better recognition effect. The data set is relabeled and divided into 10 categories according to the numbers 0-9. The new data set is trained, tested and verified using the FCOS¹ algorithm.

4. Algorithm evaluation and analysis

4.1. The experimental environment

The experimental environment of this article is: Inter® Core™ i5-10210U CPU @ 3.60 GHz, 16 GB memory, Windows 10, 64-bit system. The programming language is Python, and the GPU acceleration library is CUDA9.0¹⁵ and CUDNN7.6¹⁶.

4.2. The data set

The data set for training the detection and recognition network consists of the electric energy meter data set published by the Federal University of Paraná, Brazil, the pictures of industrial instruments from the Internet, and the pictures taken in the factory environment from different time periods of the day. The data set has a total of 6000 pictures. Preprocess the image, and perform data enhancement before inputting the training data into the network to reduce overfitting. Then randomly generate training set, test set and validation set at a ratio of 8:1:1. In the training period, ResNet-50 is used as the backbone feature extraction network, the SGD optimizer is used, the initial learning rate is set to 0.01, and the batch size is set to 16.

4.3. The result of the detection

Part of the detection effect is shown in Figure 5.



Fig. 5. figure of the industrial instrument detection results.

4.4. Evaluation index

This paper compares the accuracy (Precision, P), recall (Recall, R), and F1 of the model detection after training, and is calculated by formulas (5) – (7):

$$P = \frac{TP}{TP + FP} \quad (5)$$

$$R = \frac{TP}{TP + FN} \quad (6)$$

$$F_1 = \frac{2PR}{P + R} \quad (7)$$

Among them, TP is a real case, FP is a false positive case, FN is a negative positive case. The evaluation parameter evaluation table of the model is shown in Table 1.

Table 1. Evaluation parameter table before and after model improvement.

Method	Precision/%	Recall/%	F1/%
FCOS before improvement	84.3	86.7	85.48
FCOS after improvement	87.5	91.2	89.31

4.5. Recognition result

Recognition uses the pictures after detecting the reading area as the training set. During the training period, the learning rate is artificially modified. When epoch: [1-135], learning rate is 0.1; epoch: [136-185], learning rate is 0.01; epoch: [186-240], learning rate is 0.001. Since there will be multiple digital display areas in a picture, the number of test sets for digital recognition is 2000. Here we are not concerned about the confidence level, but only whether the recognition result is correct. The recognition effect diagram is shown in Figure 6. The accuracy of the experimental results is shown in Table 2.

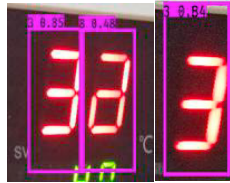


Fig. 6. figure of the industrial instrument recognition results.

Table 2. Recognition accuracy rate.

Data set	Amount	Precision/%
Test set	2000	91.05

Due to the small changes in sample morphology, the main reasons for individual number recognition errors include light, water mist, blur, and appearance jumping. In the next work, the recognition accuracy will be further optimized. The recognition results of some wrong numbers are shown in Table 3.

Table 3. Some wrong number recognition results.

Figure	Experimental results	Right/Wrong
	8	Wrong
	5	Wrong

5. Conclusions

This paper uses an improved FCOS algorithm to detect and recognize the digital display area of the industrial instruments in the workshop. Specifically, based on the FCOS basic network model, the Balanced Feature Pyramid is combined at the FPN to further integrate feature information and enhance the original features. The Softer-NMS algorithm is used to replace the traditional non-maximum suppression algorithm to improve detection accuracy. Constructing a network model, after training, testing and verification, achieved good detection efficiency and accuracy. In future research, we will try to further improve the detection accuracy and speed to meet the requirements of industrial production with high precision and real-time.

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